

Temporary bypass

9% CR

M01F026

PREFLASH CRUDE
OUT LET

#2

M01B039

PREFLASH CRUDE
OUTLET

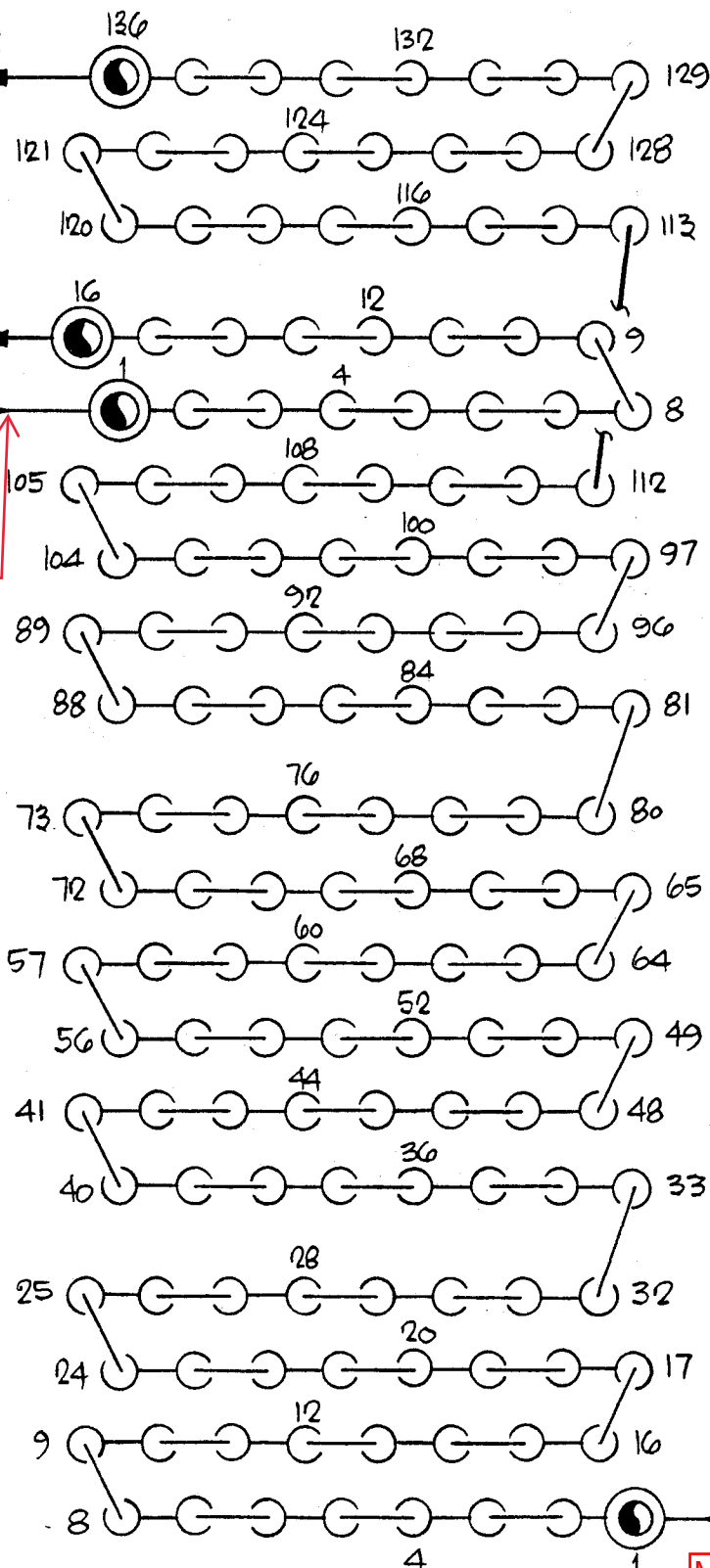
#1

PREFLASH CRUDE
INLET

#1

M01B038

Decoke Out
4"-300#



Decoke In
4"-300#

M01B040

NORTH END LOOKING SOUTH

CONVECTION SECTION (B-102)

070151

B-151

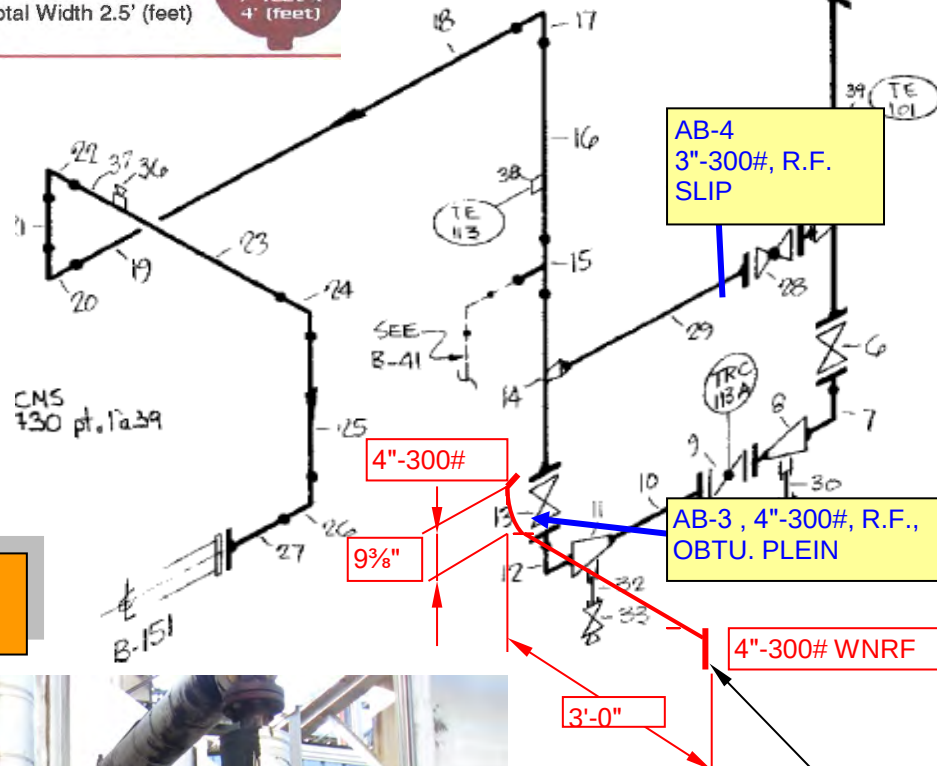
(0-1762-4"-3PI)



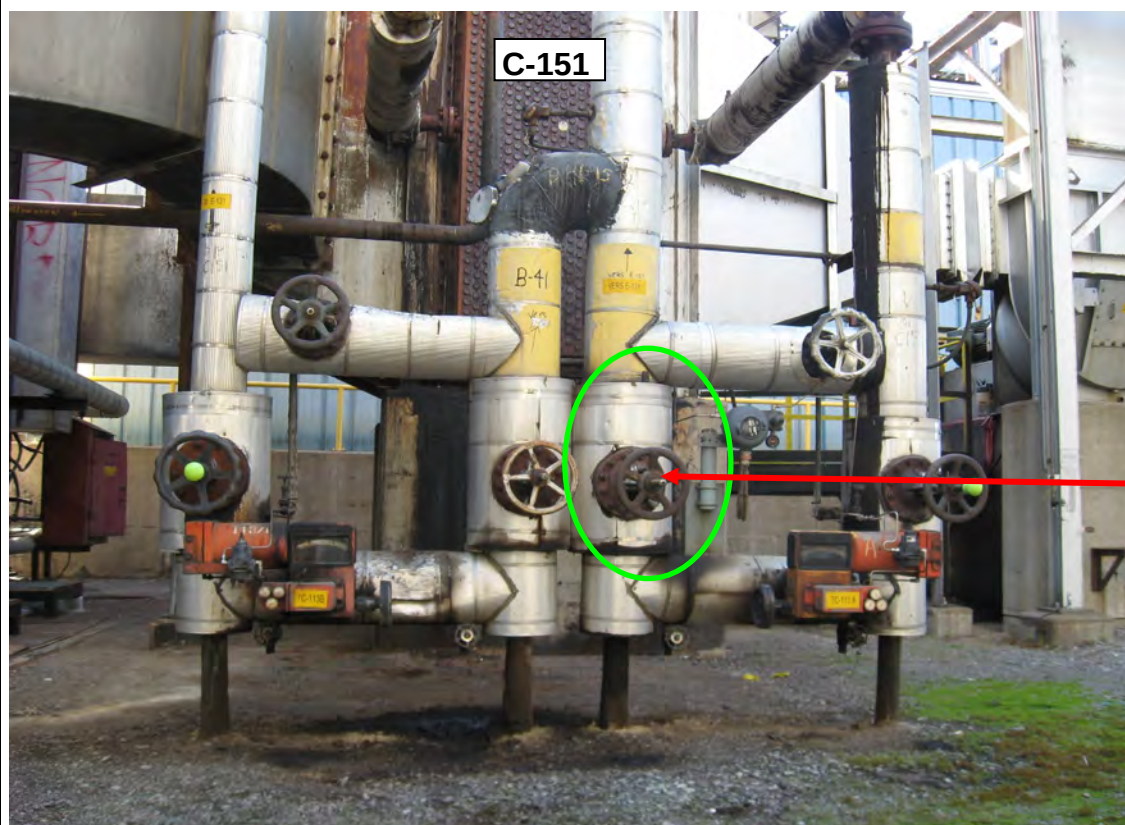
4" (inch) Launcher / Receiver Total Length 5' (feet)
300 Pound Raised Faced Flange

Total Width 2.5' (feet)

The safe working area required around this launcher would be 7' feet x 4' (feet)



Temporary piping



DÉCOKAGE B-151-MK-1

Install temporary piping
Decoke IN

Remove 4"-300#
to install temporary spool

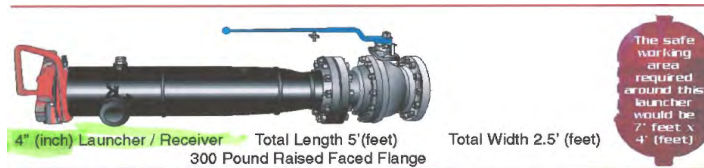
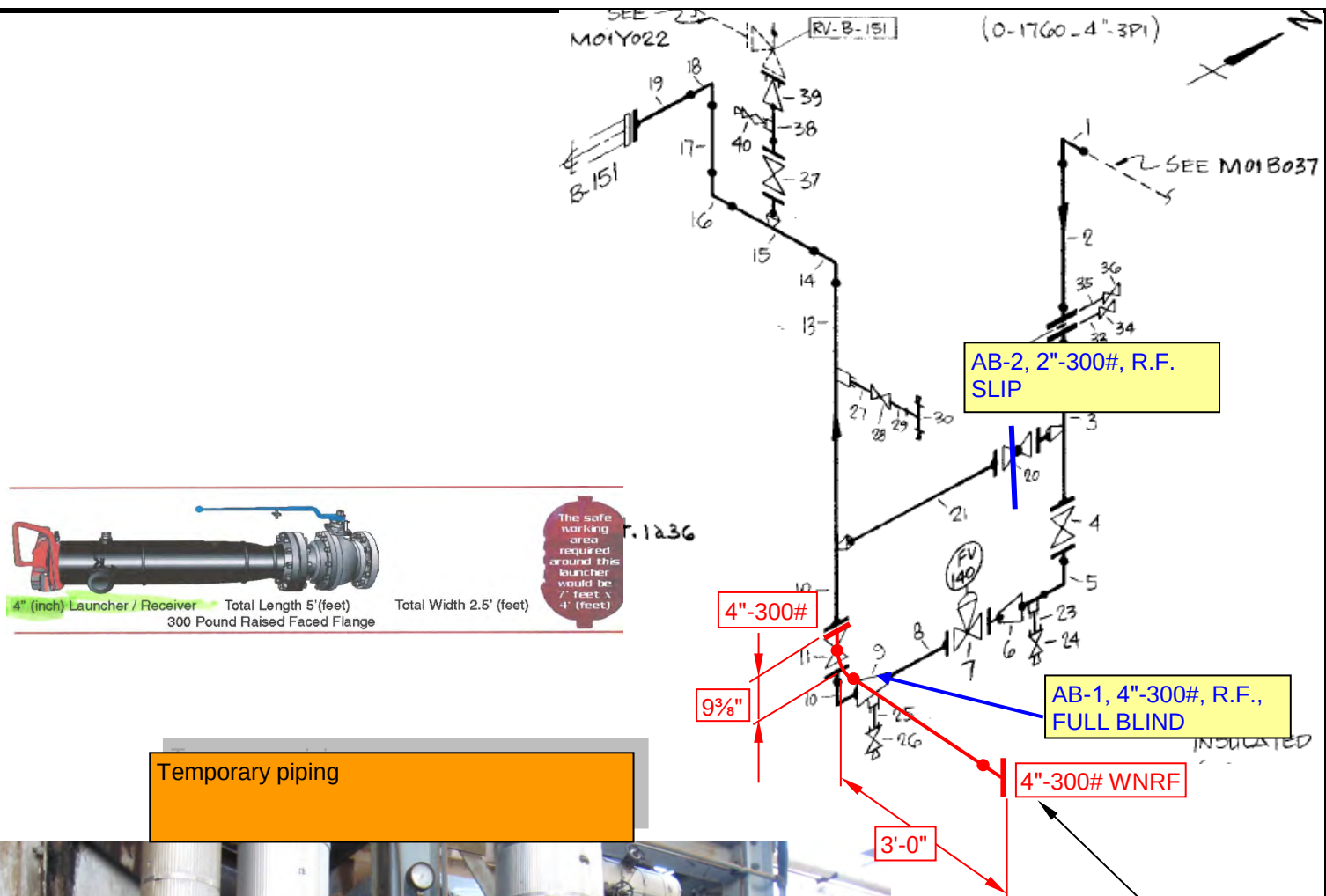
CODE DE SERVICE: B-151

DÉCOKAGE FOUR A CONVECTION - B-151

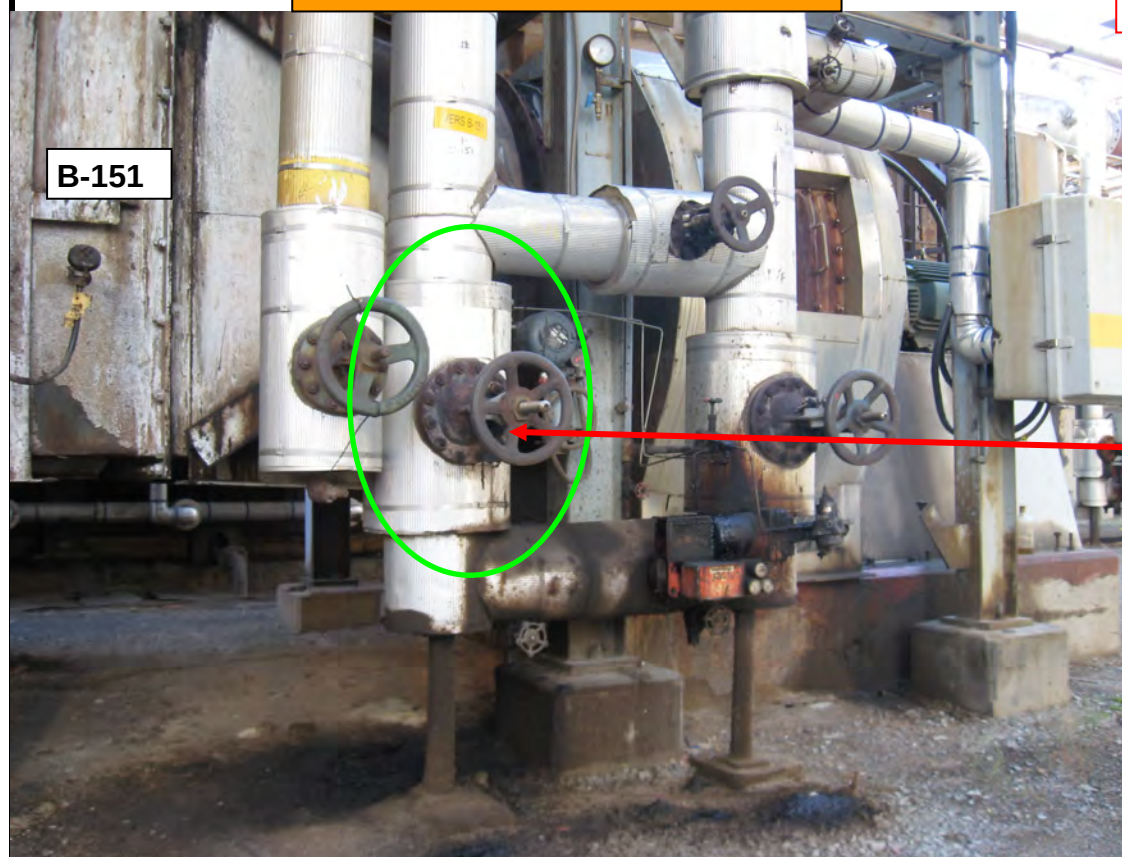
AVIS#

ORDRE#

DESCRIPTION SOMMAIRE DES TRAVAUX:



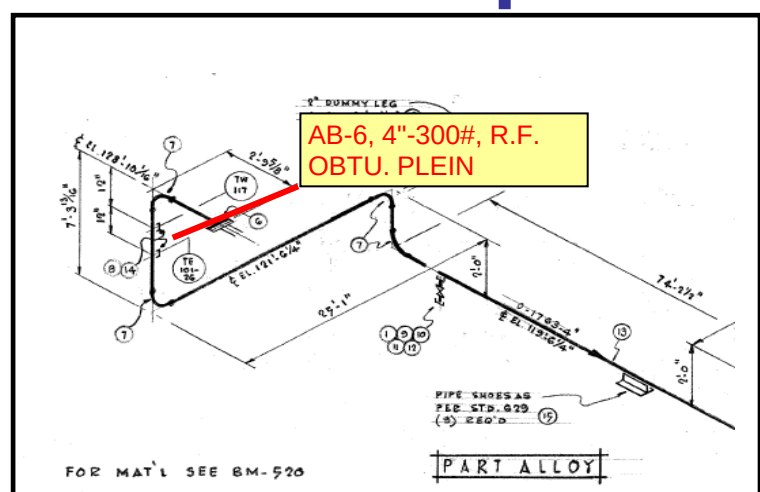
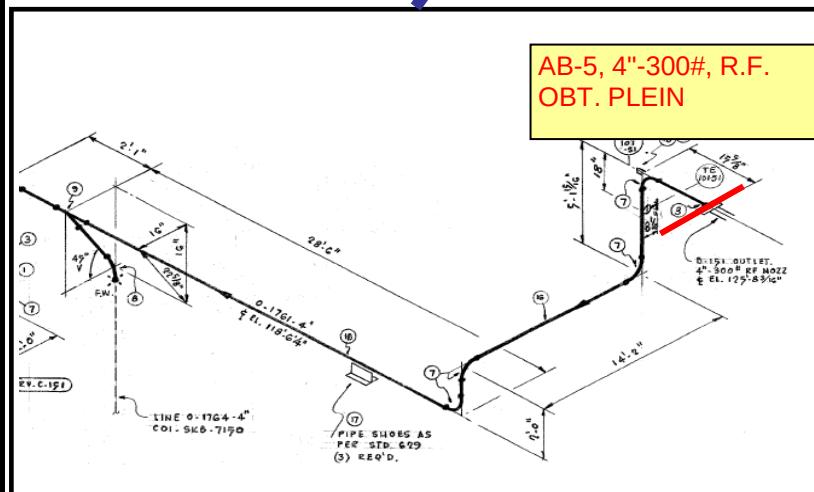
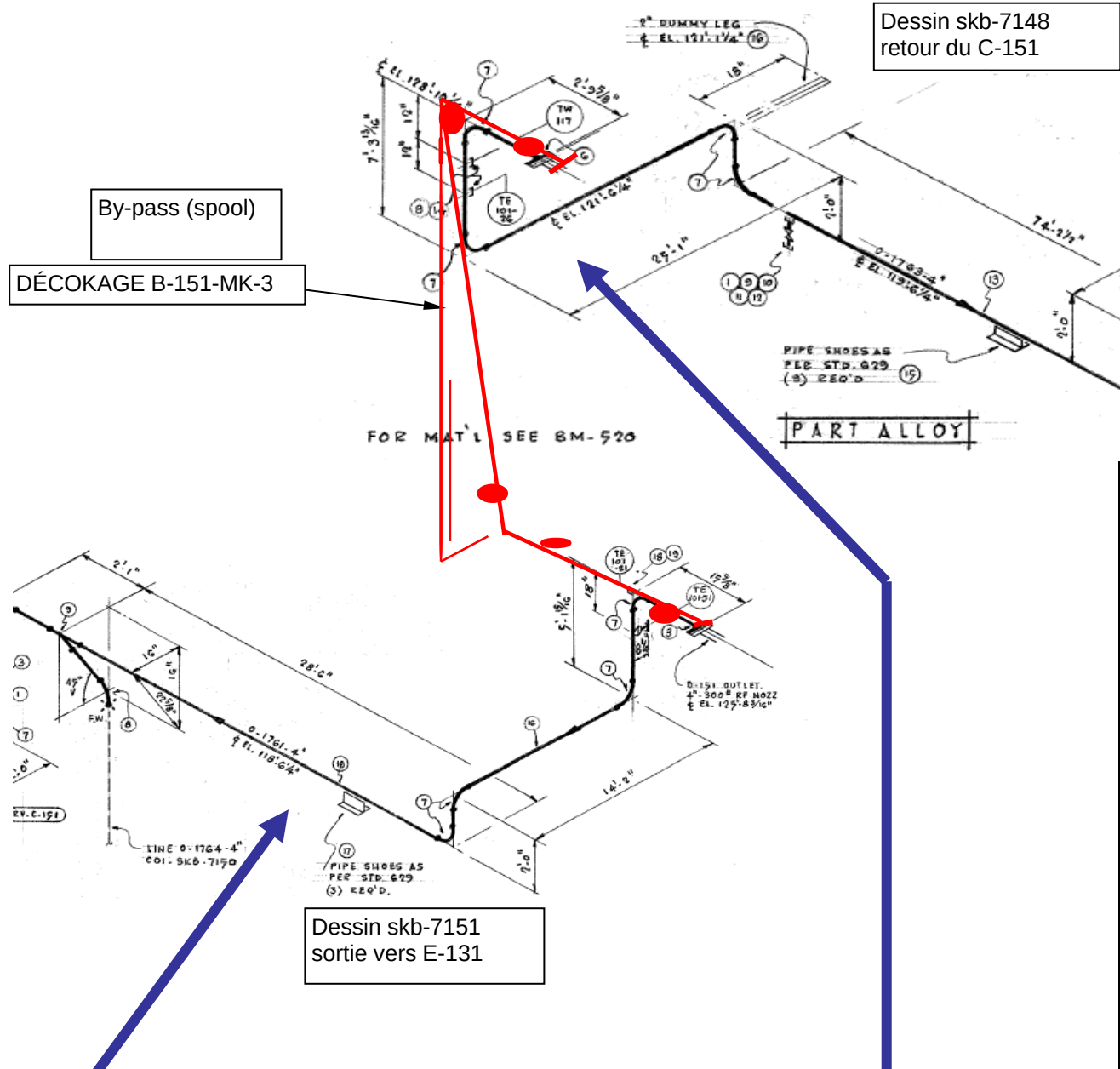
Temporary piping



DÉCOKAGE B-151-MK-2

- Install temporary piping
Decoke OUT
- Remove 4"-300#
to install temporary spool

CODE DE SERVICE: M01B038		FOUR A CONVECTION	
AVIS#		ORDRE#	
DESCRIPTION SOMMAIRE DES TRAVAUX:			
Page: 3 de 4		Rev:	00



CODE DE SERVICE: B-0151

B-151 Decoke

AVIS#

ORDRE#

DESCRIPTION SOMMAIRE DES TRAVAUX:

NOZZLE SUMMARY & INSTRUMENT CONN'S			
SYMBOL	QTY	SERVICE	SIZE, FACING & MAT'L
(IN P)	2	PROCESS INLET	4" 300# R.F.W.N. FLG ASTM-A181 Gr. I (SCH 40 BORE)
(OL P)	2	PROCESS OUTLET	4" 300# R.F.W.N. FLG ASTM-A181 Gr. I (SCH 40 BORE)
(D HB)	2	HEADER BOX DRAIN	2" 3000# FS COUPLING (CS) SCREWED (PLUGGED) HRS-1-535
(SS HB)	12	STEAM SNUFF CONN HEADER BOX	1/2" 3000# FS COUPLING (CS) SCREWED (PLUGGED) HRS-1-545
(TE)	5	TEMPERATURE ELEMENT	P.C.L. STD. B22
(DG)	3	DRAFT GAGE	P.C.L. STD. B13 DETAIL "A"

TUBE DESCRIPTION						FINNING		
MKN#	QTY	O.D.	WALL	LENGTH	MATERIAL	BARE "A"	FIN LENGTH	BARE "B"
CFT-1	14-6	4.5"	0.237 AW	22'-3"	ASTM-A106 Gr. B	4"	2'-7"	4"
CFT-2	1	4.5"	0.237 AW	25'-6"	do	3'-7 1/2"	2'-7"	4"
CFT-3	3	4.5"	0.237 AW	24'-9"	do	2'-10 1/2"	2'-7"	4"
CFT-4	2	4.5"	0.237 AW	24'-0"	do	2'-1"	2'-7"	4"
CJO-1	1	4.5"	0.237 AW	6'-0" (CUT TO SUIT)	do			
CJO-2	1	4.5"	0.237 AW	3'-0" (CUT TO SUIT)	do			

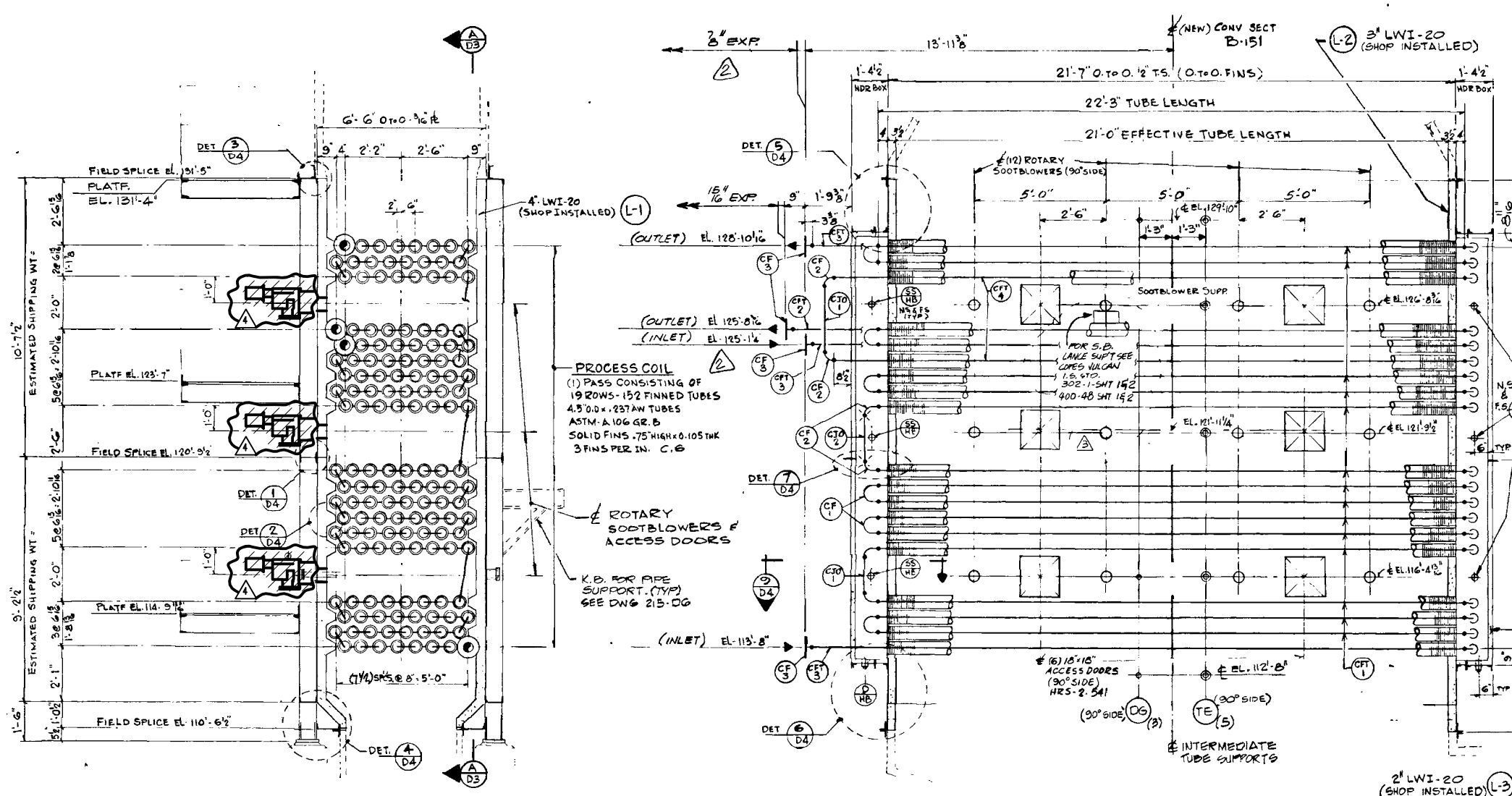
FITTING DESCRIPTION				
MKN#	QTY	SIZE & TYPE	MATERIAL	REMARKS
CF-1	147	4" SCH 40-180° S.Z. U-BEND	ASTM-A234 Gr. WPB	BEVELLED FOR WELDING
CF-2	6	4" SCH 40-90° SR ELBOW	do	do
CF-3	4	4" 300# R.F.W.N. FLG (TO MATCH SCH 40 PIPE)	ASTM-A181 Gr. I	do

INSULATION SCHEDULE					
SYM	LOCATION	DESCRIPTION	INSTALLED	ANCHORS	REMARKS
(L1)	CONV. WALLS	4" CASTABLE INSUL. PLIBRICO LWI-20	SHOP	SST 310	HRS-3-208 DET-1 CLIPS VC-4 HRS-1-202
(L2)	END TUBE SHEETS	3" CASTABLE INSUL. PLIBRICO LWI-20	do	do	HRS-3-208 DET-4 L-CLIPS VC-1 HRS-1-217 DET-1 VC-8 HRS-1-202
(L3)	CONV. HDR. BOX	2" CASTABLE INSUL. PLIBRICO LWI-20	do	C.S.	HRS-3-208 DET-5 CHAIN LINK
(L4)	CONV. CORBEL	8" CASTABLE INSUL. PLIBRICO LWI-20	do	SST 310	HRS-3-208 DET-2 DIM'A. B L-CLIP LC-8 HRS-1-217
(L5)	HOT FLUE GAS DUCTS	4" CASTABLE INSUL. PLIBRICO LWI-20	do	do	HRS-3-208 DET-1 CLIPS VC-4 HRS-1-202
(L6)	CONV. OUTLET DUCTS	2" CASTABLE INSUL. PLIBRICO LWI-20	do	C.S.	HRS-3-208 DET 5 CHAIN LINK
(L7)	HOT AIR DUCTS	2" CASTABLE INSUL. PLIBRICO LWI-20	do	do	do
(L8)	STACK NEW SECT.	3" KAO WOOL 6" / FT. DENSITY	do	do	1" THK BLANKET
(L9)	FIELD SPLICES	1/2" THK KAO WOOL BLANKET 6" CUFF DEN.	FIELD	-	AS REQ'D

- ### COIL FABRICATION NOTES
- All material on tube & fitting description list to be furnished by others.
 - Fabricator shall furnish all welding rods & all other materials req'd for Fabrication, Testing, & Shipping prep.
 - denotes Shop Weld
 denotes Field Weld (F.W.).
 - All bolt holes in inlet and outlet flanges shall straddle vertical centerline.
 - Pressure parts are designed in accordance with API-RP-530.
 - Fabrication, Inspection, Marking & Shipping shall be in accordance with H.R.C. Purchasing Specifications FSS-001, FSS-002 & FSS-007, Eng. Std. HRC-4-2573 (Weld Rod Selection) Eng. Std. H.R.C. 4-6871 (Weld Preparation) & Petrofina Canada Ltd. Spec. - 401. Where conflict exists Petrofina Canada Ltd. Spec. shall govern.
 - All pressure part welds shall be 100% radiographed.
 - Process coil design conditions are:
Design Tube Temperature 750°F.
Pressure 200 PSIG
Shop & Field Hydrostatic Test to 465 PSIG.
 - Convection coils are shop assembled including end tube supports & insulation. (Tubes are stabbed thru end tube sheets prior to welding U-bends in place.)
 - Dimensions from centerlines of heater to face of terminal connections (bevelled end or face of flange) must be maintained.
 - Tubes supplied cut to length & bevelled for welding. Tube supplied without bevelled ends must be bevelled for welding by coil fabricator per HRC Std. 4-6871.
 - U.A. Label required.

- ### GENERAL NOTES CONT. (STEEL)
- All dimensions and elevations on drawings have been assumed accurate. However, it is the contractor's responsibility to field check all dimensions and elevations before prefabrication and construction. The contractor shall notify the P.C.L. representative of any discrepancies found during the field check.
 - All structural steel to conform to ASTM A-36 or Alternate CSA G40.20 or G40.21.
 - All structural steel fabrication and erection to be in accordance with H15-1P77 and CSA Standards latest editions.
 - All duct dimensions are out to out of plate unless otherwise noted.
 - All ductwork is to be fabricated in sections as shown with flanged joints for field bolting. Duct fabricator to furnish bolting and 1/8" thickness asbestos gasketing tape for joints (Johns-Manville 132 or eq.) include 10% spare bolts and tape.
 - Field connections shall be bolted unless otherwise noted.
 - All bolts shall be ASTM-A-325-F (High Strength) or equal, unless noted otherwise.
 - Erection bolts shall be 3/4" with 7/8" holes, unless otherwise noted.
 - Inside of casing plate in flue gas ducts, furnace walls and floors shall be cleaned as per SSPC-SP6 and coated ~~with~~ in accordance with P.C.L. Spec 1001.
 - Support Steel & Ductwork shall be cleaned as per SSPC-SP 10 and receive one (1) shop coat of Primer ~~and~~ in accordance with HRC Spec. FSS-506.
 - Shop install temporary stiffening members at open ends of all duct sections. Members to be removed on completion of erection.
 - Requirements and scope for the installation and curing of castable lining shall be in accordance with manufacturers requirements and HRC Spec. FSS-506.
 - Insulation thickness shall be maintained through out by use of depth indicators.
 - Platform steel shall be sandblasted as per SSPC-SP10 and primed, ~~and~~
 - Inside of casing plate in Non-flue gas ducts shall be cleaned of mill scale, dirt, oil etc., in accordance with SSPC-SP2 or SSPC-SP3 for hand or power tool cleaning.

- ### EXTENT OF SHOP FABRICATION
- Convection Section shall be completely shop assembled (including insulation & tube coils) and shipped to field in two (2) sections with loose jumper assembly MK-CJO-2 for field welding.
 - Platform and ladders shall be shipped assembled in as large sections as are practical.
- ### GENERAL NOTES
- All elevations indicated are from grade elevation - 102'-0".
 - Platform elevations are to the top of supporting steel.
 - Platform shall be: 1" deep rectangular type welded steel grating, with 3/16" thick bearing bars at 1-3/16" C.C.
 - Ladders & Platforms shall be fabricated in accordance with Canadian Petrofina, Ltd. Engineering Standards: STD-141, 142, 143, 144, 145, 147, 151, 152.



SECTIONAL ELEVATION
0° END (LOOKING SOUTH)

LONGITUDINAL SECTION
270° SIDE (LOOKING EAST)

SHUTDOWN - 2004

CERTIFIED & APPROVED FOR CONSTRUCTION
BY *[Signature]* DATE 7/6/81
HEAT RESEARCH CORPORATION
NEW YORK

HEAT RESEARCH CORPORATION
NEW YORK

DRAWN BY: *[Signature]* SCALE: 1"=0' DO NOT SCALE THIS DWG.
CHECKED BY: *[Signature]*
APPROVED BY: *[Signature]* 79-606-4-A-3

NO.	DATE	REVISION	BY	REMARKS
4	07-20	ROTARY SOOTBLOWERS REMOVED. ORD. 12027778 S/D 2004	FC	PV
3	09-18	SOOT BLOWER SUPPORT REVAMP ADDED	A.C.	W
2	08-18	ADD. ADDITIONAL INFORMATION		
1	05-13	APPROVED GENERAL NOTES 5 TO 19 (STEEL)	P.P.	
0	03-20	APPROVED FOR CONSTRUCTION		

215-D4	GENERAL ARRANGEMENT
215-D2	" "
215-D1	" "
DWG NO.	TITLE
REFERENCE DRAWINGS	
PETROFINA CANADA LTD. LTEE. POINTE-AUX-TREMBLES, P.Q. REFINING DEPARTMENT	
EFFICIENCY IMPROVEMENT OF CRUDE HEATER B-102	
GENERAL ARRANGEMENT B-151 CONVECTION SECTION ELEVATIONS	
DRAWN	101010P
CHECKED	312-81
DATE	4-25-79
UNIT	1
CLASS	B
DWG NO.	215-D3
REV.	4